‘येथे बहुतांचे हित ।’

Web Development Project
Partial PBL Review

Department of Information Technology
A.Y. 2025-26 Sem-II

Subject: Web Development

Class: IT-1 **Batch: C**

Team Members:

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Project Domain

Domain Name: Hotel Management System

Many hotels face difficulties in managing reservations, customer records, room availability, and billing through manual processes. This often leads to errors, delays, and poor customer service. There is a need for a digital system that can efficiently manage room bookings, customer details, payments, and hotel services in a centralized and organized manner.

Target User :

1) Hotel managers and staff
2) Hotel guests/customers
3) Receptionists and front desk operators
4) Hotel administrators

Objective :

1. To provide an easy-to-use system for managing hotel operations.
2. To simplify room booking and reservation processes.
3. To maintain accurate records of customers and room availability.
4. To ensure secure and efficient billing and payment management.

Website Overview

Brief Description of the Website:

The Hotel Management System is a web-based application that allows users to check room availability, make reservations, and manage their bookings online. It also helps hotel staff manage room details, customer information, and billing efficiently through a centralized system.

Key Features Implemented:

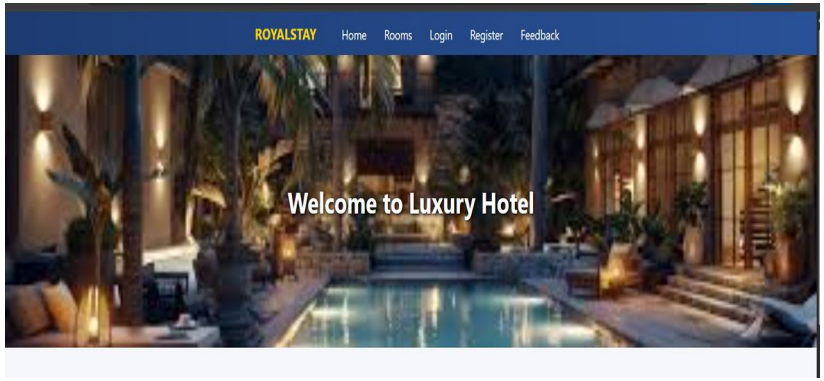
1) user Registration & Login
2) Room Availability Display
3) Room Booking / Reservation System
4) Customer Details Management
5) Payment Integration (basic/demo)
6) Booking History
7) Admin Panel (optional)

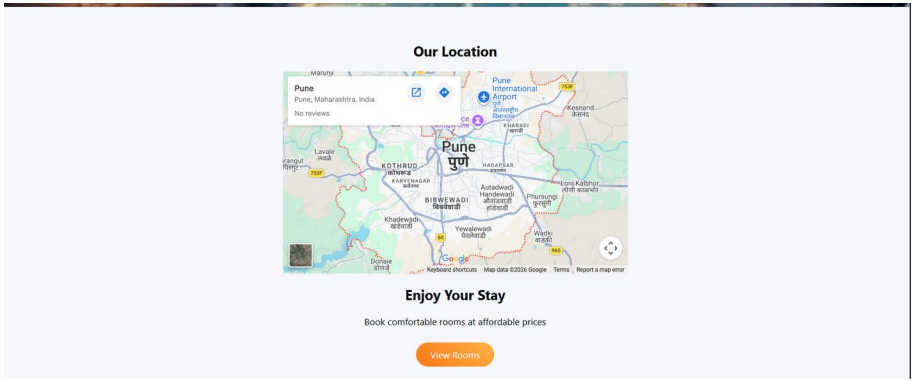
Technology Stack Used:

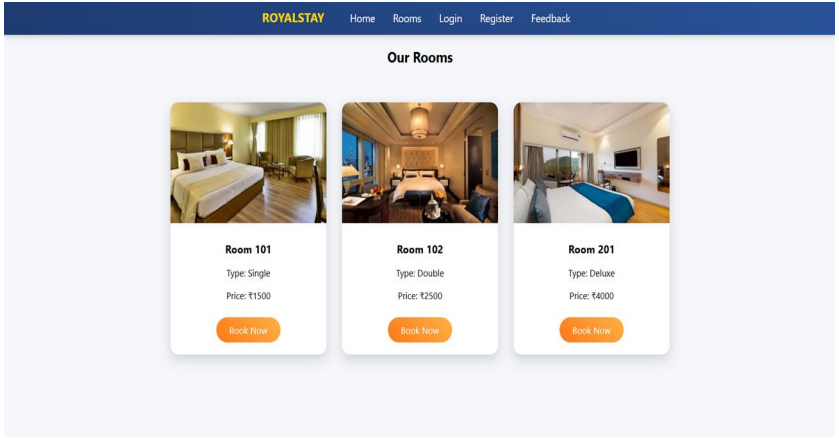
1) HTML – Structure of web pages
2) CSS – Styling and layout
3) JavaScript – Interactivity and validation

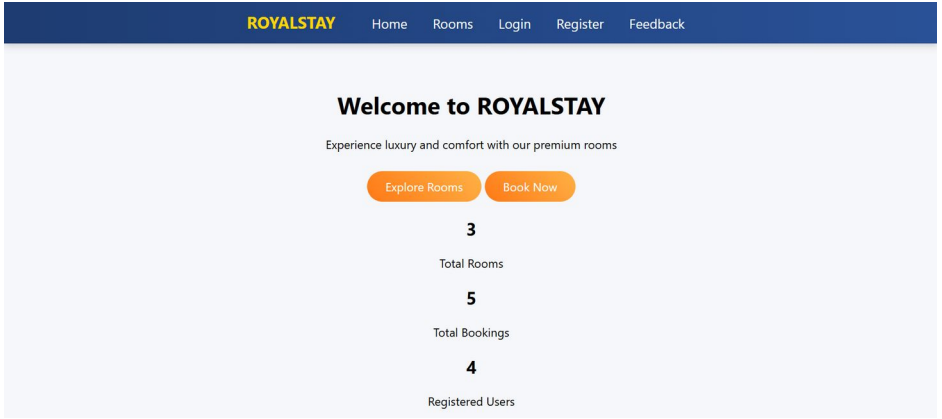
Complete UI Design

Homepage Screenshot

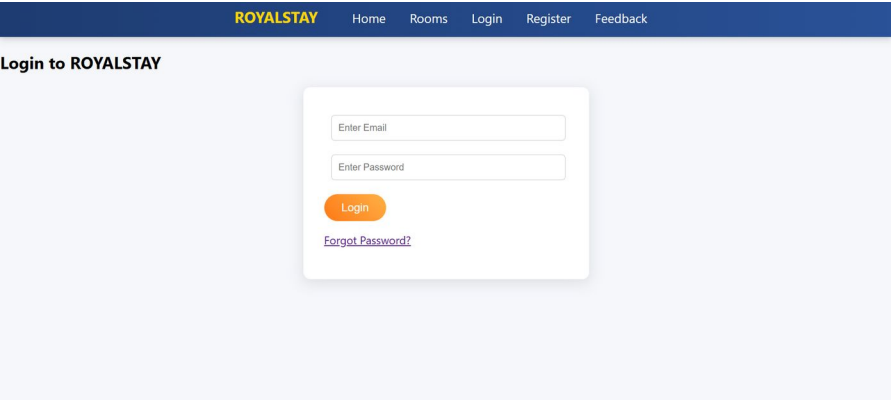


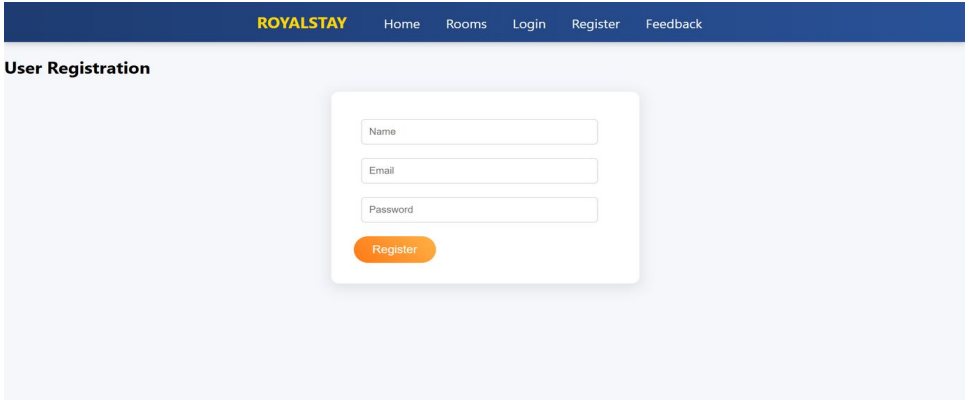


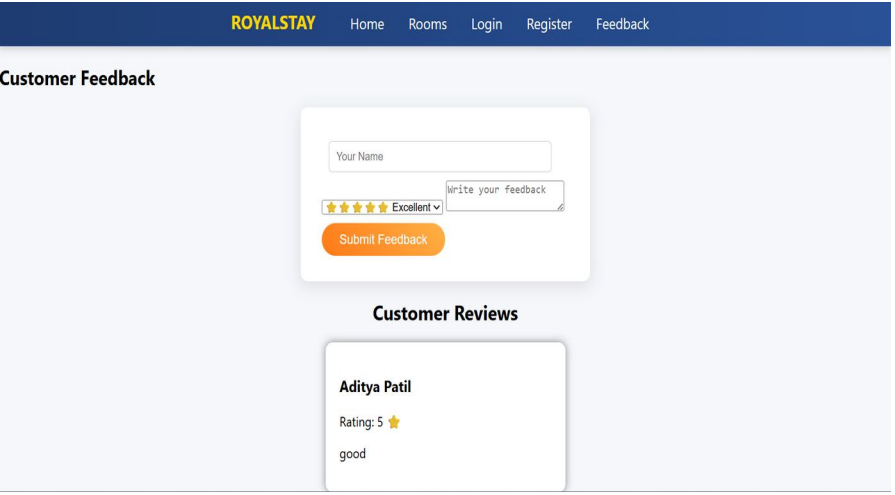


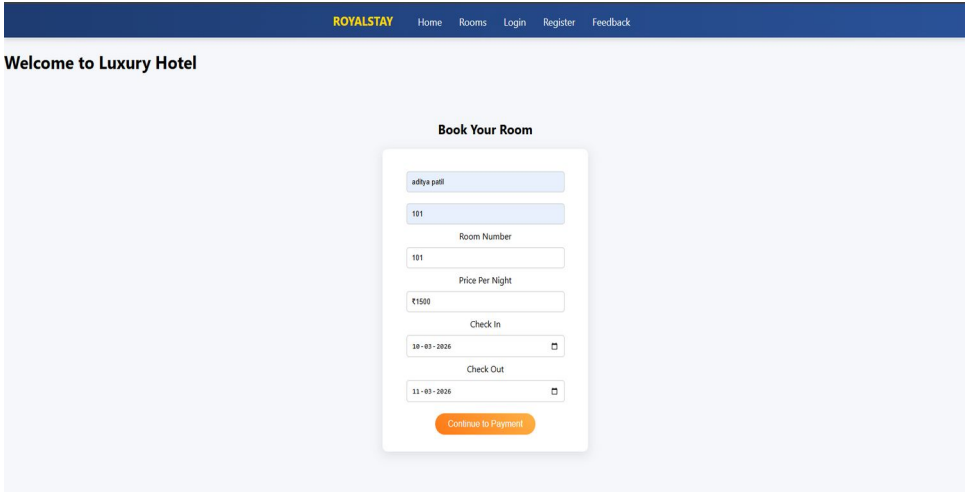


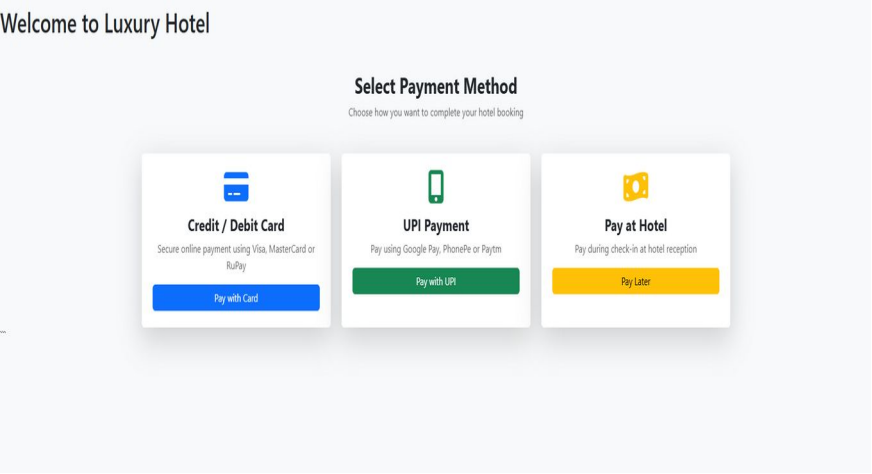
Internal Pages Screenshots

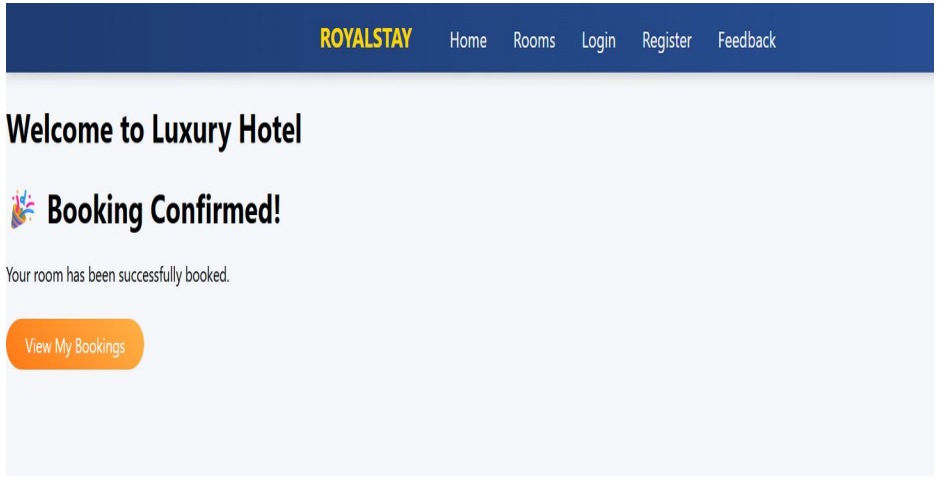


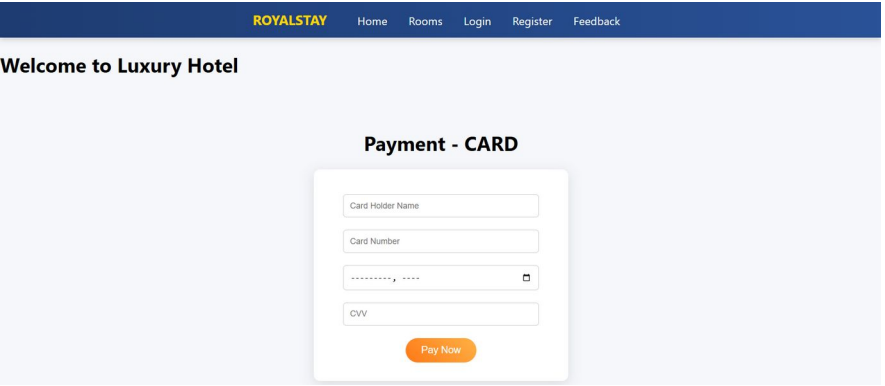


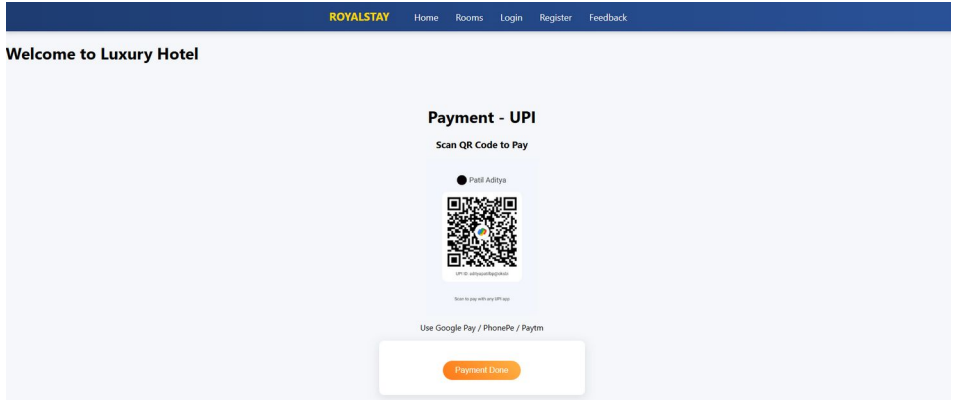





Welcome to Luxury Hotel
Select Payment Method
Choose how you want to complete your hotel booking
Credit / Debit Card
Secure online payment using Visa, MasterCard or RuPay
Pay with Card
UPI Payment
Pay using Google Pay, PhonePe or Paytm
Pay with UPI
Pay at Hotel
Pay during check-in at hotel reception
Pay Later


ROYALSTAY
Home
Rooms
Login
Register
Feedback
Welcome to Luxury Hotel
Booking Confirmed!
Your room has been successfully booked.
View My Bookings


ROYALSTAY
Home
Rooms
Login
Register
Feedback
Welcome to Luxury Hotel
Payment - CARD
Card Holder Name
Card Number
CVV
Pay Now


ROYALSTAY
Home
Rooms
Login
Register
Feedback
Welcome to Luxury Hotel
Payment - UPI
Scan QR Code to Pay
Patil Aditya
Use Google Pay / PhonePe / Paytm
Payment Done

ROYALSTAY Home Rooms Login Register Feedback

Welcome to Luxury Hotel

Payment - CASH

Pay at Hotel

You can pay at the hotel during check-in.

Confirm Booking

Booking Summary

Room: 101

Price per night: ₹1500

Total Days: 2

Total Amount: ₹3000

Responsive Design Feature

1. Mobile-friendly layout using CSS media queries
2. Flexible grid system (Flexbox / Grid)
3. Responsive images
4. Adjusted font sizes for smaller screens
5. Navigation adapts for mobile view

Navigation Pane & Page Flow

Navbar / Sidebar Design

Page Linking Structure

User Flow Explanation

Home → Main landing page

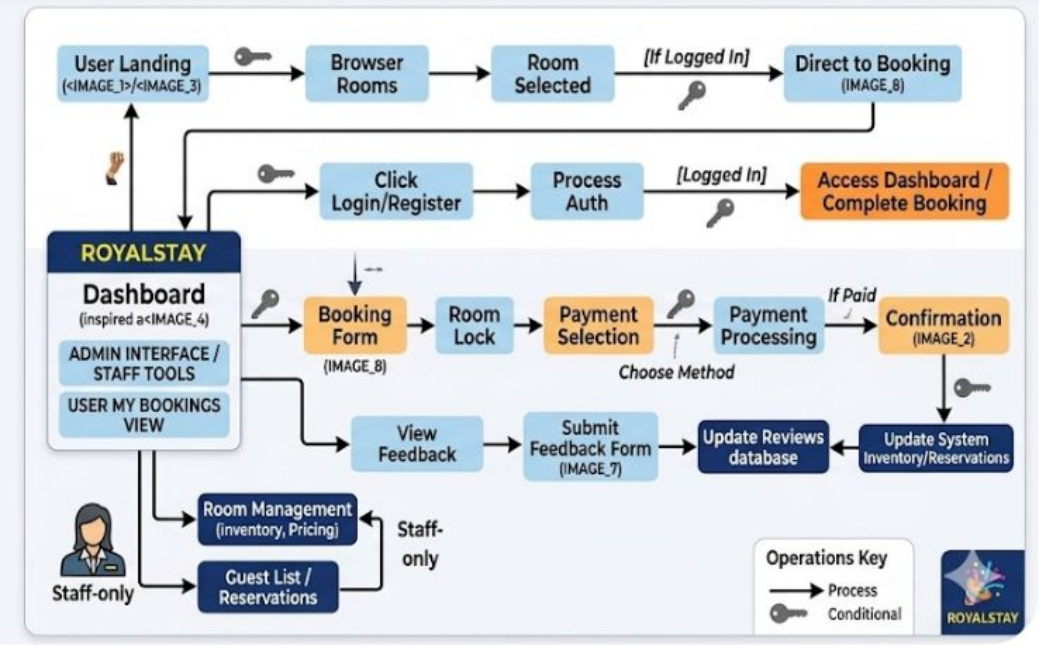
About → Information about the pet hotel

Team → Staff information

Services → Grooming, Boarding, Care services

Blog → Pet care articles

Appointment → Booking form page



Easy page is direct accessible from the main navigation bar

JavaScript Validation (Each Page)

Form Validations Implemented

1) Email/Password Validation Rule

Required Field – Email cannot be empty.

Valid Format Required – Must follow standard email format:
example@domain.com

2) **Must Contain:**

One @ symbol
Valid domain name (e.g., gmail.com)
At least one . after domain

3) **Error Handling Message**

General Form Errors

❌ "Please fill all required fields.“

❌ "Something went wrong. Please try again later.“

Success Message

✔ "Thank you for your feedback!"

Database Table Design

1. List of Tables
   1. Customers
   2. Rooms
   3. Bookings
   4. Feedback

1.Customers

Customer_ID (Primary Key)

Name

Phone

Email

2.Rooms

Room_ID (Primary Key)

Room_Number

Price

Status

3.Bookings

Booking_ID (Primary Key)
Customer_ID (Foreign Key)
Room_ID (Foreign Key)
Check_In_Date
Check_Out_Date
User_Name
Payment_Status

4.Feedback

ID (Primary Key)
Booking_ID (Foreign Key)
Name
Rating
Message

5.Relationship

a.Customer → Bookings

One Customer can make many Bookings

Relationship: **1 : N**

b. Room → Bookings

One Room can be reserved in many Bookings

Relationship: **1 : N**

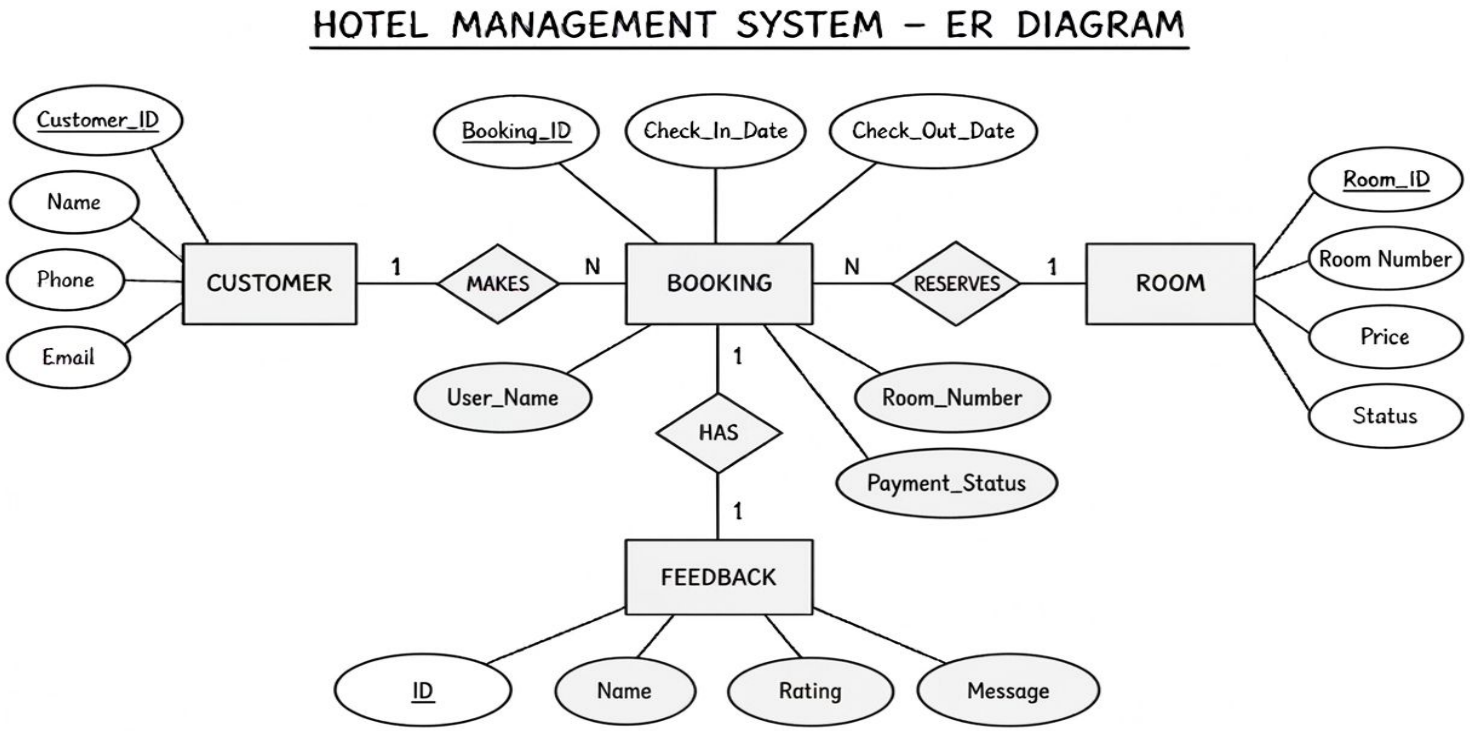
c. Booking → Feedback

One Booking has one Feedback

Relationship: **1 : 1**

Entity Relationship Diagram

Insert ER Diagram Image here



1. Customer

This entity stores information about customers who book rooms.

Attributes:

customer_id (Primary Key) – unique customer ID

name – name of the customer

phone – contact number

email – email address

2. Room

This entity stores details of rooms available in the hotel.

Attributes:

room_id (Primary Key) – unique room ID

room_number – number of the room

price – cost of the room

status – availability status (available, booked, etc.)

3. Booking

This entity stores information about room bookings made by customers.

Attributes:

booking_id (Primary Key) – unique booking ID
check_in_date – check-in date
check_out_date – check-out date
user_name – name of the person booking
payment_status – payment status (paid, pending)
customer_id (Foreign Key) – identifies the customer who made the booking
room_id (Foreign Key) – identifies the room booked

4. Feedback

This entity stores feedback given by customers after their stay.

Attributes:

id (Primary Key) – unique feedback ID
name – name of the customer
rating – rating given by customer
message – feedback message
booking_id (Foreign Key) – identifies the booking related to feedback

Conclusion

Project Status

1.Challenges Faced

The **Hotel Management System** project has been successfully developed to simplify hotel operations such as room booking, customer management, and reservation handling. The system allows users to check room availability, register or log in, and book rooms online in an easy and efficient way. It helps hotel staff manage customer information and room details digitally, reducing manual work and improving accuracy. During the development of the project, some challenges were faced such as managing room availability to avoid double booking, designing a simple and user-friendly interface, and implementing proper booking validation. Despite these challenges, the system was successfully implemented using web technologies.

2.Future Enhancements

In the future, the system can be improved by adding several advanced features. Online payment gateway integration such as **UPI, Razorpay, or PayPal** can be implemented to allow secure online payments during booking. A real-time room availability tracking system can be added to improve booking efficiency. A complete admin dashboard can be developed for managing rooms, customers, and reservations more effectively. Notifications through email or SMS can be implemented to inform users about booking confirmations and updates. The user interface can also be improved with better design, animations, and advanced JavaScript features. Additionally, a mobile application version of the system can be developed to make hotel booking more convenient for users.